

Special Review

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Index terms:

Molecular analysis
Radiology and radiologists, research
Review

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Abbreviations:

FHPG = 9-[(3-fluoro-1-hydroxy-2-propoxy)methyl]guanine
FMAU = 2'-fluoro-5-methyl-1-beta-D-arabinofuranosyluracil
HSV-Tk = herpes simplex virus 1 thymidine kinase
NIH = National Institutes of Health

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Molecular Imaging¹

The term molecular imaging can be broadly defined as the in vivo characterization and measurement of biologic processes at the cellular and molecular level. In contradistinction to “classical” diagnostic imaging, it sets forth to probe the molecular abnormalities that are the basis of disease rather than to image the end effects of these molecular alterations. While the underlying biology represents a new arena for many radiologists, concomitant efforts such as development of novel agents, signal amplification strategies, and imaging technologies clearly dovetail with prior research efforts of our specialty. Radiologists will play a leading role in directing developments of this embryonic but burgeoning field. This article presents some recent developments in molecular sciences and medicine and shows how imaging can be used, at least experimentally, to assess specific molecular targets. In the future, specific imaging of such targets will allow earlier detection and characterization of disease, earlier and direct molecular assessment of treatment effects, and a more fundamental understanding of the disease process.

Advances in molecular and cell biology techniques, the ability to decode entire genomes, the continuous search for new targets, and the unraveling of the molecular pathways of many diseases have had a marked effect on the way we practice medicine today. Much research attention has been rightfully directed toward understanding the cellular and molecular mechanisms of diseases, but efforts have also been directed toward the development of noninvasive, high-resolution, in vivo imaging technology. Specifically, over the past 2 years, in vivo molecular imaging has been identified by the National Cancer Institute as an extraordinary opportunity for studying diseases noninvasively and, in many cases, quantitatively at the molecular level. Although there still remains a scientific gulf between the basic scientists who discover new genes and their function and the imaging scientists who could transform these discoveries into noninvasive imaging methods, this gap is rapidly closing. With substantial funding now available in this new field, investigators from diverse disciplines have been attracted and important inroads into molecular imaging have been made. Although it is unlikely that many presently practicing radiologists will engage in the field “hands on” in the near future, it is imperative that they keep abreast of recent developments, help recruit the next generation of clinician scientists, and support the transition of novel methods into clinical practice.

Molecular imaging is a growing research discipline aimed at developing and testing novel tools, reagents, and methods to image specific molecular pathways in vivo, particularly those that are key targets in disease processes. While certain imaging techniques that may be defined as “molecular” were developed decades ago (eg, imaging with monoclonal antibodies or receptor imaging with nuclear techniques), it is only recently that a host of needed adjunct basic research tools have become routinely available. Some of these tools that we can now build on include molecular cloning, microfabrication, chip arrays, robots, x-ray crystallography, fast mass spectrometry, and sophisticated computer analysis. These more recently available basic science tools now allow us to answer basic biologic questions in vivo and to do this in a high-throughput fashion.

It is expected that the fruits of today's molecular imaging research will have a direct effect on patient care within the next 5–15 years. Our current assessment of disease is based on anatomic changes or, more recently in specialized cases, physiologic changes that are a late manifestation of the molecular changes that truly underlie disease. Direct imaging of these molecular changes will directly affect patient care by allowing much earlier detection of disease. We may potentially be able to image molecular changes that we

TABLE 1
Glossary of Commonly Used Terms

Term	Definition
Amplification	Molecular biology: increase in the number of copies of specific DNA fragment (polymerase chain reaction); molecular imaging: increase in imageable signal
Angiogenesis	Growth of new blood vessels
Apoptosis	Programmed cell death involving a regulated intracellular proteolytic enzyme cascade
Base pair	Two bases (adenine and thymine or guanine and cytosine) held together by weak bonds
Bioinformatics	Use of computers for the creation and query of extensive databases on genomes, protein sequences, etc
cDNA	Complementary DNA that is synthesized from an mRNA template
Cloning	1: Production of cells, all genetically identical, from a single ancestor; 2: production of multiple copies of a single gene or DNA segment
DNA	Deoxyribonucleic acid, the molecule that encodes genetic information
Gene	Ordered sequence of nucleotides, located in a certain position on a chromosome.
Gene expression	Process by which a gene's coded information is converted into proteins and other structures operating a cell
Gene product	RNA or protein resulting from expression of a gene
Gene therapy	Treatment or prevention of disease by transferring genes
Genome	Sum of all the genetic information in chromosomes of a given organism
Hybridization	Reversible binding of two complimentary strands of DNA or DNA/RNA to form a double stranded molecule
Kilobase	Unit of DNA length equaling 1,000 nucleotides
Knockout animal	Animal with a mutation or deletion of a specific gene
mRNA	Messenger RNA, which serves as a template for protein synthesis
Mutation	Change in DNA sequence
Nucleotide	Subunit of DNA or RNA consisting of a base (adenine, guanine, thymine, or cytosine in DNA; adenine, guanine, uracil, or cytosine in RNA), a phosphate, and a sugar molecule (deoxyribose in DNA and ribose in RNA)
Oncogene	A gene associated with cancer; many oncogenes are involved in controlling the rate of cell growth
Plasmid	Autonomously replicating circular extrachromosomal DNA distinct from the bacterial genome; some plasmids are capable of integrating into the genome; multiple genetically engineered plasmids are also used as cloning vectors
Polymerase chain reaction	Amplification of a DNA sequence by using a polymerase and specific base primers; successive cycles result in rapid and specific amplification of the desired sequence
Progenitor cell	Immature cells that are capable of self-renewal and differentiation (eg, hematopoietic progenitor cells can differentiate into red cells, white cells, and platelets)
Promoter	Site on DNA to which RNA polymerase will bind to initiate transcription
Protein	Large molecule composed of amino acids in a specific sequence; proteins are essential for cellular functioning
Reporter gene	Encodes for an easily detectable protein (lac z for histology or imaging marker genes for in vivo imaging; see Table 2)
RNA	Ribonucleic acid; classes of RNA: messenger RNA, transfer RNA, ribosomal RNA, other small RNAs
Sequencing	Determination of the order of nucleotides in DNA or RNA; sometimes also refers to extracting the order of amino acids in proteins
Stem cell	Earliest progenitor cell from which differentiated cell types originate
Transcription	Production of RNA from DNA template
Transduction	Successful gene transfer and expression (viral system)
Transfection	Successful gene transfer and expression (nonviral system)
Transgene	Foreign gene that is transferred into a target cell or tissue
Transgenic animal	Organism that contains gene from another organism put into its genome
Translation	Production of proteins from RNA template
Vector	Vehicle (virus, liposome) in which a gene can be delivered
Virus	Noncellular biologic entity that can only reproduce in host cells

currently define as “predisease states,” which would allow intervention at a time when the outcome is most likely to be affected. In addition, by directly imaging the underlying alterations of disease, we will potentially be able to directly image the effects of therapy. Thus, we will play a direct role in determining the effectiveness of treatment shortly after therapy has been initiated, in contradistinction to the many months often required today to determine whether pharmacologic or biologic intervention has been beneficial.

This review is intended as a short primer on the subject matter for the uninitiated. Because the field may represent an entire “new world” to many readers, we have included a short glossary of commonly used terms (Table 1). Specifically, in this review we will (a) discuss how best

to image specific molecular targets; (b) review some of the basics aspects of DNA, genes, and the Human Genome Project; (c) review imaging efforts in gene therapy; (d) use angiogenesis and apoptosis as specific examples on how to image at the molecular level; and (e) review recent advances in phenotypic imaging of transgenic and knockout mice. Wherever possible, we provide more specific background information for further in-depth reading.

WHAT AND HOW TO IMAGE

To image specific molecules in vivo, several key criteria must generally be met (Fig 1): (a) availability of high-affinity probes with reasonable pharmacodynamics; (b) the ability of these

probes to overcome biologic delivery barriers (vascular, interstitial, cell membrane); (c) use of amplification strategies (chemical or biologic); and (d) availability of sensitive, fast, high-resolution imaging techniques. In a typical scenario, all four prerequisites must be met for successful in vivo imaging at the molecular level.

Target identification and validation with high-affinity probes is one of the key prerequisites for interrogation of specific molecular targets in living systems. Such probes can be small molecules—for example, receptor ligands or enzyme substrates. Alternatively, higher molecular weight affinity ligands (“biotechnology drugs”) are often utilized (eg, monoclonal antibodies, recombinant proteins). Although the design of potential affinity ligands against thousands of targets may

appear daunting at first glance, recent advances in drug discovery technology (combinatorial techniques, rational design, high-throughput testing, robotics, target identification, and validation through genomic sciences) have helped move this process forward rapidly. For example, the data points (to describe the effect of one compound on one target) generated by large screening programs in industry were roughly 200,000 in the early 1990s, 5 million in the mid-1990s, and over 50 million today. Many such "hits" (ie, compounds that elicit desired effects or properties) are subsequently validated in more complex biologic systems. Current therapeutic drugs are directed against approximately 500 molecular targets (45% of which are receptors; 30%, enzymes; and 25%, other targets). Estimates of potential future drug targets are in excess of 5,000–10,000. In other words, there are at least 10 times more targets that will be discovered and exploited than are being used today. While the use of these powerful methods have essentially changed therapeutic drug development, there has been limited impact on the design of imaging probes to date. There are many reasons for this, primarily the smaller market size for imaging drugs as compared with that for therapeutic drugs and the lack of active interactions between imaging and chemistry departments. As imaging receives further recognition as a key enabling technology for in vivo molecular target assessment, such essential interdisciplinary interactions will increase, undoubtedly leading to the discovery of many novel imaging agents.

High-affinity ligands developed rationally, combinatorially, or by chance must have the ability to reach the intended target at sufficient concentration and for a sufficient length of time to be detectable in vivo. Rapid excretion, non-specific binding, metabolism, and delivery barriers all counteract this process and must be overcome. Delivery barriers are typically the most challenging to deal with, particularly for larger biotechnology drugs. However, even low-molecular-weight agents may not be easily internalized into cells, a requirement for imaging of intracellular targets. A number of strategies have been developed to circumvent existing delivery barriers, while current research is concentrated on the development of even more efficient methods. Examples include the use of peptide-derived membrane translocation signals that result in active shuttling of imaging drugs into cells (1), pegylation to de-

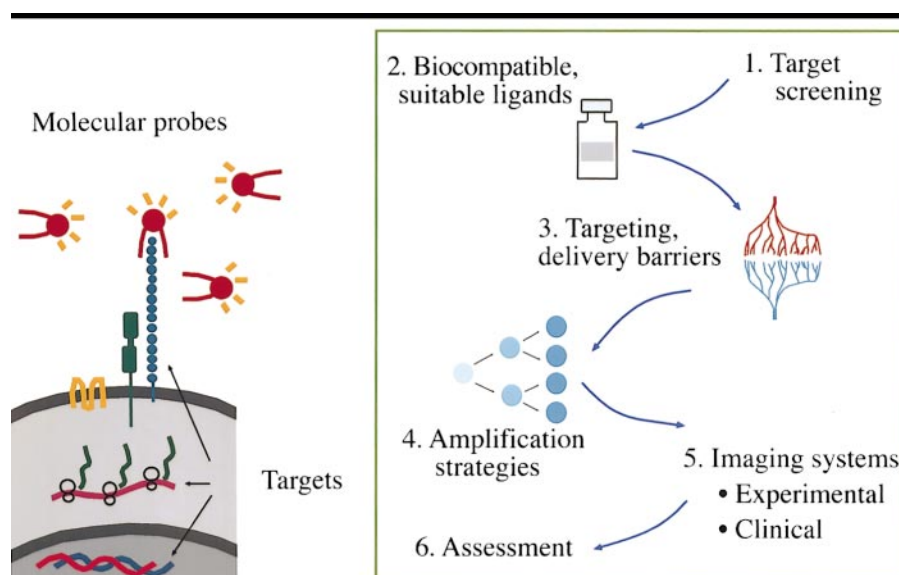


Figure 1. Schematics show prerequisites to in vivo molecular imaging. Potential targets can be at the DNA, RNA, or protein level (also see the text and Fig 2).

crease both immunogenicity and rapid recognition (2), use of long-circulating drugs to achieve a more homogenous distribution (3), and/or local delivery combined with pharmacologic or physical methods to improve targeting (4,5). Another aspect unique to in vivo molecular imaging is the frequent inability to eliminate unbound affinity ligands, which may markedly contribute to background noise. In in vitro assays, this step is easily dealt with by washing off unbound ligands and then recording specific signals, with resultant high target-to-background ratios. Unfortunately, in the in vivo situation, on the contrary, options are limited to optimization of pharmacokinetics (ie, waiting for nonspecific agents to be washed out). This may be aided by removing some of the ligand from circulation (into the reticuloendothelial system) by means of the addition of specific "chase" compounds shortly before imaging (6).

Utilization and future development of efficient amplification strategies (increasing the imageable signal) remain critical components of much of molecular imaging research. Figure 3 summarizes generic cellular targets, including DNA, messenger RNA (mRNA), and proteins. Although the numbers of DNA and mRNA targets per cell are limited (requiring extreme levels of signal amplification for visualization), the imaging of proteins and/or protein function is much more feasible (referred to as "imaging downstream"). A number of chemical and biologic amplification strategies have been developed to facil-

itate molecular imaging at the protein level: These strategies make use of (a) improving target concentration (by means of pretargeting [7], avidin-biotin amplification systems [8], and improving kinetics [9]), (b) unique cellular functions (trapping of converted ligands [10,11]), or (c) the ability of probes to change their physical behavior after target interactions (fluorescence dequenching [12], increasing R2 [13] or R1 [14] at magnetic resonance [MR] imaging). The use of some or all of the strategies, together with the selection of a downstream target of gene expression, will typically suffice for imaging at the protein level.

Molecular information can be obtained with some but not all of our presently used "high-end" imaging technologies (Fig 3). Despite recent advances in imaging technology, further improvements in current imaging modalities and exploration of new modalities are still at the center stage of molecular imaging research. For example, the development of optical imaging technology (including diffuse optical tomography, phase-array detection, photon counting, near-infrared fluorescence imaging), high-spatial-resolution MR and nuclear imaging techniques (eg, micro-MR, micro-positron emission tomography [PET]) play an important role in the field (discussed later in this article). Improved spatial resolution now allows imaging of mouse models of human disease, and imaging findings and concepts can thus be directly translated into a clinical context.

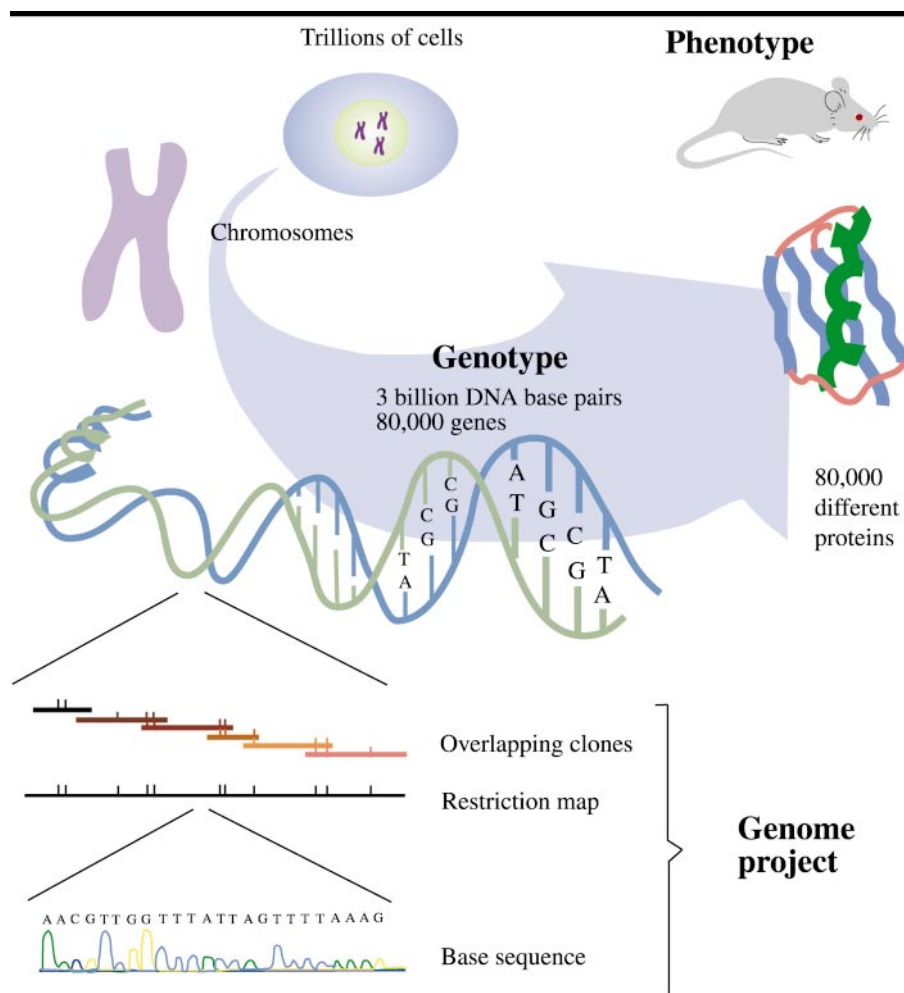


Figure 2. Schematic shows genome, DNA, genotype, and phenotype. Mammalian cells contain multiple chromosomes (46 in humans, 40 in mice, 42 in rats). In the human, the chromosomes are made up of roughly 3 billion base pairs that encode approximately 80,000 genes. The goal of the Human Genome Project is to identify the exact sequence of these base pairs. Individual genes encode for proteins essential for the function of cells and phenotype. A = adenine, T = thymine, C = cytosine, G = guanine.

DNA, GENES, AND THE HUMAN GENOME PROJECT

The entire body of genetic information required to form and sustain life is contained in the DNA molecule (Fig 2). Most living entities, including viruses and bacteria, fungi, plants, animals, and humans use this universal instruction language. DNA is composed of four chemical bases: adenine, thymine, cytosine, and guanine. Strands of these bases pair with each other (adenine and thymine, cytosine and guanine) to form a structure that resembles a ladder (Fig 2). The entire ladder is compacted around other proteins (involved in maintaining structure and regulating gene expression) and is referred to as a chromosome. Collectively, the 23 chromosome pairs in a hu-

man cell are referred to as the human genome.

A single gene typically consists of several thousand base pairs. There are an estimated 60,000–100,000 genes in the human genome. When a gene becomes active, the DNA strands separate, and a new molecule is synthesized by using the DNA as a template. This new molecule, RNA, acts as a template for the final products: proteins. The estimated 80,000 proteins perform all functions of life inside and outside cells.

Variations in DNA that predispose to illness may exist from birth (genetic defects) or may be acquired during the course of life (mutation). Although DNA will frequently copy itself and perform protein synthesis countless times, occasional mistakes in DNA replication or

damage to DNA do occur. On rare occasions, the cellular machinery is unable to repair or the body is unable to remove these mistakes, and, as a result, disease occurs—for example, cancer. Knowledge of the sequence of the human genome allows the identification and understanding of the function of all human genes. From here, one can begin to unravel the complicated processes of biology.

The Human Genome Project is a 13-year effort coordinated by the U.S. Department of Energy and the National Institutes of Health (NIH) to decipher the specific sequence of all human genes. The project, which started in 1990, currently draws over \$300 million in funding annually. A rough draft of the sequence of the entire genome was reported to be completed in June 2000 (15,16). It is expected that the price for decoding the entire human genome will ultimately approach \$4 billion. The specific goals of the Human Genome Project are to (a) identify the approximately 80,000 genes in human DNA; (b) determine the exact sequence of the 3 billion base pairs that make up human DNA; (c) store this information in databases; (d) develop faster and more efficient sequencing technologies; (e) develop tools for data analysis; and (f) address ethical, legal, and social issues arising from the project. Apart from the human genome project, the decoding of many other genomes has already been completed (eg, *Escherichia coli*, *Mycoplasma pneumoniae*, *Haemophilus influenza*, *Saccharomyces cerevisiae*, *Helicobacter pylori*), while other projects are currently underway (eg, the mouse genome; www.ornl.gov/hgmis/faq/compngen.html#completegenomes).

The impact of the Human Genome Project is expected to revolutionize medical practice and biologic research well into the 21st century. For the first time, we will have the specific codes that govern life (the equivalent to having the source code of a computer program, or better, of all computer programs). All of the human genes will eventually be found, and accurate diagnostics will be developed for most of the inherited diseases. Mouse models of human disease will be more easily developed, facilitating the understanding of gene function in health and disease. We can also use the information to specifically screen the genome for disease risks. By using the blueprints, we can start to develop specific therapies (rational drug design) targeted at specific proteins (eg, protease inhibitors, receptor agonists). We will also be

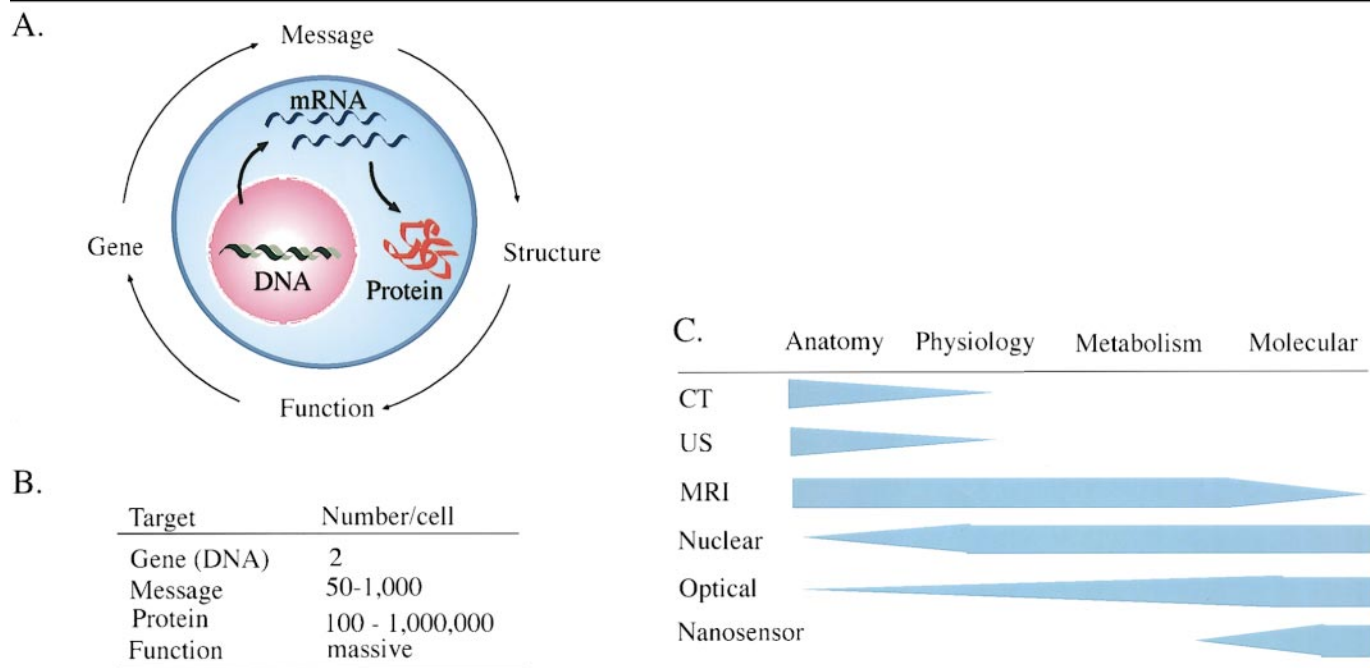


Figure 3. Schematics show molecular imaging targets. *A*, Different intracellular imaging targets and their function are summarized. *B*, Numbers of targets per cell. *C*, Use of current imaging technology for molecular imaging. Nanosensor refers to the use of semiconductor microfabrication technology to create sensors sensitive to biologic change. (Adapted, with permission, from reference 111.)

able to manipulate genes directly by transferring new or missing genes (gene therapy). Furthermore, custom drugs will become available (pharmacogenomics) through such research efforts and screening. In addition to these benefits, anticipated benefits of genome research will become apparent in fields of microbial genomics (biologically derived fuels, toxic waste clean-up), risk assessment (radiation and chemical risks, mutations), evolution, and DNA forensics (DNA matching, organ donor matching) and agriculture (disease-resistant crops, biologically derived pesticides, vaccines), among others.

The role of imaging and the expertise of our specialty in these efforts and applications have been largely unexplored until recently. Yet, the influence of the Human Genome Project on diagnostic imaging will be widespread, both in experimental research and in clinical applications. Identification of new targets for earlier disease detection, evaluation of specific molecular markers for therapy assessment, use of imaging for drug screening, and imaging of gene expression are just a few of the more immediate apparent applications. The following sections will describe in more detail the basics of imaging of gene expression and highlight some specific examples.

Disease		Vectors	
Cancer	72%	Retrovirus	53 %
Infectious	8%	Adenovirus	19 %
Monogenic	16%	Poxvirus	6 %
CV	3%	AAV	1 %
Other	1%	HSV	< 1%
Technique		Naked DNA	5 %
In vivo	48%	Lipids	13 %
Ex vivo	52%	Other	< 1%

Figure 4. Schematic shows clinical gene therapy trials broken down by disease, technique, and vector system used. AAV = adeno-associated virus, CV = cardiovascular, HSV = herpes simplex virus.

CLINICAL GENE THERAPY TRIALS

Gene therapy has been heralded as a potential revolution in medicine because, for the first time, a therapy is aimed at correcting the cause of disease rather than treating phenotypic symptoms. Gene therapy also represents a platform technology that can be applied to a wide range of diseases and targets. In an ex vivo gene therapy approach, cells are removed from the patient, treated, and then introduced again. In the in vivo ap-

proach, the gene (in the form of a vector) is directly administered to the patient. More than 4,000 patients have been enrolled in over 400 clinical gene therapy studies over the past decade. Approximately two-thirds of trials are directed at cancer, with the remainder directed at AIDS, cystic fibrosis, and a few other diseases (Fig 4).

While early clinical trials with gene therapy were heralded as a major breakthrough in modern medicine, the outcomes of trials in the middle and late 1990s have generated limited enthusi-

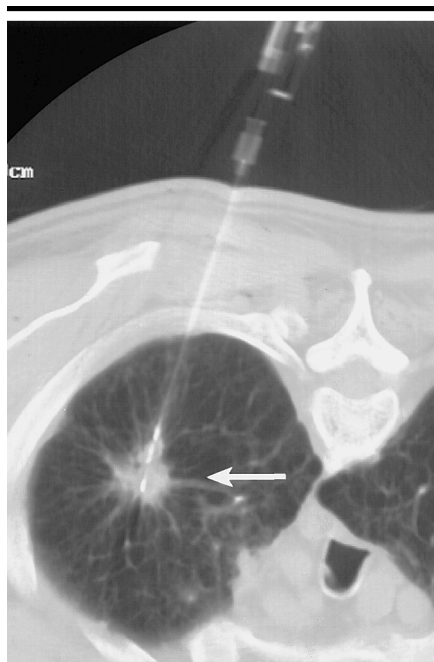


Figure 5. Image-guided delivery of an adenoviral vector encoding for p53 tumor suppressor gene; p53 is also known as the “guardian of the genome” and is mutated or deleted in a large number of malignancies. Transverse CT scan of the lung (patient is prone) demonstrates a lung cancer (arrow) into which the viral vector was administered coaxially through a guiding needle. (Image courtesy of Stephen Swisher, MD, and Marshal Hicks, MD, University of Texas M.D. Anderson Cancer Center, Houston.)

asm and even some backlash (17). This reflects, in part, the high safety profiles (conservative dosing) of early clinical trials, inefficient vector systems, and/or unreasonably high expectations of the community. It has become clear that ex vivo and in vivo gene therapy are infinitely more complex than was initially hoped; an expert team from the NIH recently recommended that main resources should be directed toward adequate vector development and more appropriate preclinical testing prior to premature clinical trials (18). The current areas of focus in vector development include (a) increasing the specificity and ability of viruses to target viruses; (b) improving the duration of gene expression; (c) decreasing immunogenicity and toxicity (17); (d) improving manufacturing; and (e) establishing objective endpoints, including imaging, with which therapy can be judged. Most recently, important strides have been made by using DNA to correct inherited severe combined immunodeficiency (19). Similar data have been generated for cardiovascular and cancer

gene therapy, and, in general, the outlook for gene therapy is now much brighter than it was only 3 years ago.

Genes are typically delivered by means of either viral (75%) or nonviral (25%) delivery systems, termed vectors. Retroviral vectors are the most commonly used vectors for ex vivo gene transfer, followed by adenovirus, adeno-associated virus, and herpes simplex virus. A retrovirus transfects by integrating its genes into the chromosome of a targeted cell, a process restricted to dividing cells. An adenovirus transfers genes to both quiescent and dividing cells and does not integrate into the target cell chromosome. In general, viral vectors result in high transfection efficiency. Their main limitations are immunogenicity, difficulties in manufacturing, and, often, size limitations of DNA inserts. Nonviral systems commonly include naked DNA, DNA encapsulated in liposomes or lipids, or charged complex DNA. Nonviral systems are much less efficient at transfection, as compared with viral systems, but are less immunogenic, are easier to manufacture, and have no size limitations for DNA inserts. Overall, there currently is no single vector with generic utility for all diseases, organs, and/or gene therapy applications. What remains clear is that recent improvements in the manufacturing process for adenoviruses, retroviruses, adeno-associated viruses, and herpes simplex viruses have generated the need for in vivo imaging of both vector and target distribution, as well as in vivo imaging of the resultant transgene expression.

IMAGING GENE DELIVERY

It is anticipated that interventional radiologists will play a major role in delivering genetic material to patients, particularly as gene therapy methods become everyday clinical tools. For gene therapy vectors to work, they must be delivered efficiently to the intended target. Although systemic intravenous administration of a vector is occasionally performed, more common application strategies include stereotactic and focal image-guided applications. For cardiovascular delivery, special catheter systems have been designed to maximize interaction between the vector and the endothelium (20–22). More recently, additional interventional strategies have been used to improve gene delivery, including application of a “gene gun” (23), electroporation (24), radiation (25), high-frequency ultrasonography (US) (26,27), and other mechanical

or chemical disruptions that improve local delivery.

Figure 5 shows one example of tumoral administration of an adenoviral vector encoding for the p53 tumor suppressor gene. The vector is administered through a stereotactically placed needle, similar to needles used for biopsy or radio-frequency ablation. In this specific example, the p53 gene encodes a protein that inhibits tumor formation. In the inherited p53-deficiency form, Li-Fraumeni syndrome, young adults develop several independent tumors. In most other forms of cancer, mutations of p53 are also found, and they contribute to the complex network of molecular events leading to tumors. Thus, replacement of the mutated or missing p53 gene with a new p53 copy is one form of gene therapy against tumors. Other antitumor gene therapy approaches include introduction of pro-drug activation enzymes (28), oncolytic viruses (29,30), adoptive cellular immunotherapy (31), other forms of apoptosis induction (32), and blocking of the cell cycle (33), among others.

For the direct assessment of local and systemic distributions of a herpes simplex virus type 2 vector, hrR3, we have developed techniques for radiolabeling it with indium 111 (^{111}In)-oxine (34). The blood half-life of the virus in rats is 1 minute for the fast component (10% contribution) and 180 minutes for the slow component (90% contribution). With an intracarotid injection, the total amount of virus that accumulated in tumor was only $0.10\% \pm 0.07$ (SD) of the injected dose per gram of tissue, while the distribution to nontarget organs was much higher: liver, $27\% \pm 2.9$; lung, $2.1\% \pm 0.7$; and kidney, $1.7\% \pm 1.6$, with lesser amounts in other organs (35). When virus was injected directly into the tumor, 71% of virus remained in the tumor at 24 hours with the following distribution regions: tumor greater than border zone, which was greater than normal brain (ratio, 99:40:1). Localization and quantification of viral accumulation in vivo will enable detailed analysis of viral and organ interactions critical for advancing the therapeutic use of the ever increasing number of developed vector systems.

IMAGING OF EXOGENOUS MARKER GENES

Given the availability of animal models and vectors, ease of molecular cloning techniques, and relevance to clinical ap-

plications, the field of gene expression imaging has recently boomed. A moderate number of imaging marker genes (ie, gene products that can be detected with different imaging modalities) have been described (36). The choice of a specific system depends on the imaging requirements (single or repeated), intended use (animal or human), spatial requirements (organs vs higher resolution) and several other factors. Several excellent radiology-oriented review articles are already available on this subject (22,37,38), so the following section is limited to some specific examples that cover the different imaging modalities.

Nuclear Imaging

The basic concept underlying nuclear imaging (PET, single photon emission computed tomographic [SPECT], planar imaging) techniques is that an expressed protein can be probed for with specific radiopharmaceuticals. Two classes of such imaging marker genes have been investigated: (a) marker genes that encode for intracellular enzymes and (b) marker genes that encode for cell-surface proteins or receptors. The major advantages of intracellular protein expression are the relatively uncomplicated expression strategy and lack of recognition of the expression product by the immune system. The major advantages of surface-expressed receptors and acceptors are favorable kinetics (sometimes avoiding the need for the tracer to penetrate into a cell) and the fact that synthetic receptors can be engineered to recognize already approved imaging drugs (pertechnetate) (39).

Herpes simplex virus 1 thymidine kinase (HSV-Tk) has been used as a key pro-drug-converting enzyme for a number of anticancer gene therapy approaches (40,41). The enzyme has a broad substrate specificity and can convert less toxic ganciclovir into toxic compounds that result in cell death. Imaging of HSV-Tk expression is reliant on the use of iodinated or fluorinated (fluorine 18 [^{18}F]) prodrugs such as fialuridine (42), ganciclovir (11), penciclovir (43), 9-[(3-fluoro-1-hydroxy-2-propoxy)methyl]guanine (FHPG) (44), and 2'-fluoro-5-methyl-1-beta-D-arabinofuranosyluracil (FMAU) (45). These imaging labels freely traverse the plasma membrane by means of active transport. When HSV-Tk is present in cells, the substrates are phosphorylated by the viral thymidine kinase (but not by the human thymidine kinase) and thus become "trapped." Cellular retention of radioactivity is, therefore, an indicator of thymidine

kinase presence and thus transfection (Fig 6). Results from a number of experimental studies have validated the use of HSV-Tk, and review articles are available for more in-depth reading (46). More recently, clinical trials of HSV-Tk imaging have been initiated in Europe. Cytosine deaminase is another intracellular prodrug-converting enzyme harnessed for imaging of gene expression (47,48).

A number of other reporter systems are currently under development for imaging of gene expression by means of nuclear techniques (Table 2). These systems include naturally occurring receptors (eg, dopamine-2 receptor [58], somatostatin receptor [57], gastrin-releasing-peptide receptor [55]), naturally occurring channels (eg, iodine transporter channel [59]), other intracellular enzymes (eg, cytosine deami-

nase, which converts 5-fluorocytosine into toxic 5-fluorouracil), or artificial cell-surface-expressed fusion proteins that recognize commercially available imaging drugs such as technetium 99m ($^{99\text{m}}\text{Tc}$) diethylenetriaminepentaacetic acid, or $^{99\text{m}}\text{Tc}$ -DTPA, and $^{99\text{m}}\text{Tc}$ pertechnetate. The latter approach is detailed in more depth in a recent review article (62).

MR Imaging

A main advantage of using MR to image gene expression is its high spatial resolution and the ability to extract more than one measurement parameter at a given imaging session. Because of the natural insensitivity of MR imaging for label detection, robust cellular amplification strategies are needed to confer ade-

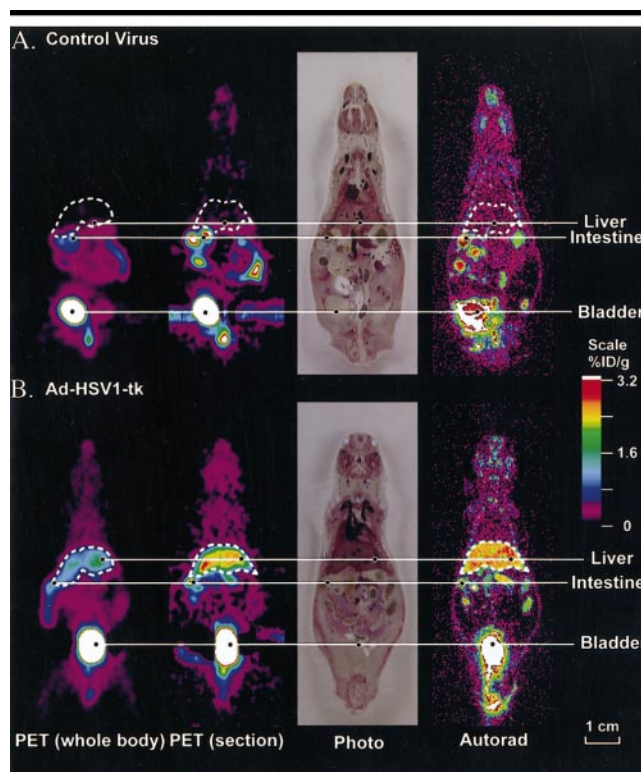


Figure 6. Images in mice demonstrate differences in thymidine kinase gene expression after tail vein injection of a replication-deficient adenovirus that expresses thymidine kinase (*Ad-HSV1-tk*) or control virus. %ID/g = percentage injected dose per gram. *A*, Injection of 1.53×10^9 plaque-forming units of control virus. *B*, Injection of 1.53×10^9 plaque-forming units of the replication-deficient adenovirus. For each mouse, a whole-body mean coronal projection PET scan (left) of the ^{18}F activity distribution was obtained. The location of the liver (dotted white outline) was determined from both the 8- ^{18}F -fluoroganciclovir signal and the cryostat slices (second from right). Coronal micro-PET sections (second from left) are approximately 2-mm thick. After PET, the mice were sectioned (second from right), and autoradiography (*Autorad*) was performed (right). Images are displayed on the same quantitative color scale to allow signal intensity comparisons among them. (Reprinted, with permission, from reference 11.)

TABLE 2
Imaging Marker Genes

Marker Protein*	Imaging Modality	Ligand or Substrate†	Mechanism
Intracellular			
Thymidine kinase (11,42–45)	Nuclear	FIAU, ganciclovir, penciclovir, FHPG, FMAU	Phosphorylation of prodrug
Cytosine deaminase (48)	Nuclear, MR spectroscopy	Cytosine, fluorinated prodrugs	Deamination of prodrug, fluorine imaging
Tyrosinase (49)	Nuclear, MR imaging	Tyrosine, dopa	Oxidation of substrates into melanin or metal scavenging
Arginine kinase (50)	MR spectroscopy	ATP and arginine	Conversion to phosphoarginine
Creatinine kinase (51)	MR spectroscopy	ATP and creatinine	Conversion to phosphocreatinine
β-Galactosidase (14)	MR imaging	Galactosylated chelators	Cleavage of galactose residues changes relaxivity
Green fluorescent protein (52)	Optical	None required	Fluorescence
Luciferase (53)	Optical	Luciferin	Bioluminescence
Proteases such as cathepsin D (54)	Optical	Quenched near-infrared-fluorescent fluorochromes	Fluorescence activation
Cell surface			
Gastrin-releasing-peptide receptor (51,140)	Nuclear	Bombesin	Affinity binding
Somatostatin receptor (57)	Nuclear	Peptides	Affinity binding
Dopamine-2 receptor (58)	Nuclear	¹⁸ F spiperone	Affinity binding
Iodine binding (59)	Nuclear	Iodine	Trapping
Fusion proteins (60)	Nuclear	^{99m} Tc chelates	Transchelation
Engineered internalizing receptor (61)	MR imaging	Transferrin	Internalization, relaxivity change

* Numbers in parentheses are reference numbers.

† ATP = adenosine triphosphate, FIAU = fialuridine.

quate sensitivity of label detection. This is usually achieved by using targeted and/or “smart” MR contrast agents coupled with biologic amplification strategies. One particularly robust amplification system is based on cellular internalization of (super)paramagnetic probes such as monocrystalline iron oxide nanoparticles (Fig 7). It has been shown that receptor expression and regulation can be visualized with MR imaging when an engineered transferrin receptor is probed with a superparamagnetic transferrin probe (13,61). These study results have provided proof of the principle that it is feasible, through the mechanism of receptor overexpression, to image receptor gene expression by using MR (13).

Another strategy to amplify genetic information for MR imaging is through a second reporter system. One such model system has recently been developed (49) and is based on the capability of a single enzyme (eg, tyrosinase) to catalyze the production of multiple other molecules (eg, melanin) in cells. In the case of the tyrosinase-melanin system, the latter becomes detectable with MR imaging (63). Tyrosinase catalyzes two fundamental reactions during melanogenesis: (a) the hydroxylation of tyrosine yielding dihy-

droxyphenylalanine, or dopa, and (b) the subsequent oxidation of dopa to dopa-quinone. The latter undergoes spontaneous cyclization and polymerization to yield melanins. Melanins have a remarkably high metal-binding capacity (up to 35% by weight), particularly for iron, which is responsible for the high signal intensity of melanotic tumors on T1-weighted MR images (63). More recently, paramagnetic substrates have been developed for tyrosinase to further amplify MR signal intensity (Bogdanov A, unpublished data, 2000).

Paramagnetic chelates that change magnetic properties at enzymatic hydrolysis (“smart contrast agents”) have recently been used for imaging of gene expression. In one example, a paramagnetic galactopyranoside/galactosidase system was injected into cells and used to image β-galactosidase activity in *Xenopus* embryos (64). Galactosidase activity was revealed by measuring enzyme-mediated changes in R1. Another enzyme/substrate system that has recently been developed is based on a different modulation of relaxation effects (Bogdanov A, oral communication, 2000). In this system, termed ELAM (enzyme-linked amplification for MR imaging), paramag-

netic products of oxidoreductase-mediated catalysis undergo spontaneous polymerization. The resulting paramagnetic polymers produce substantially greater relaxation effects than do the substrate monomers. This effect is due mainly to a change in tumbling rates of the paramagnetic moieties and results in multifold MR signal amplification.

The ability to use MR spectroscopy to distinguish signals from chemically distinct compounds also offers the potential to measure gene expression. In one study, tumor cell lines expressing complementary DNA that encodes for cytosine deaminase from yeast were grown in animals. Conversion of the nontoxic prodrug 5-fluorocytosine to the antimetabolite 5-fluorouracil by means of yeast cytosine deaminase could be observed and quantified in vivo by using fluorine 19 MR spectroscopy (47). The results of this study demonstrated local conversion of 5-fluorocytosine, validating the concept of localization of chemotherapy with enzyme-prodrug gene therapy.

Optical Imaging

A number of optical imaging approaches have recently been described,

some of which have been used for imaging of gene expression *in vivo*. The described techniques rely on fluorescence, absorption, reflectance, or bioluminescence as the source of contrast, while imaging systems can be based on diffuse optical tomography (65,66), surface-weighted imaging (reflectance diffuse tomography) (67,68), phase-array detection (69,70), confocal imaging (71–73), multiphoton imaging (74–76), or microscopic imaging with intravital microscopy (77,78). Naturally, with the exception of near-infrared fluorescence imaging (12) and superficial confocal and two-photon imaging (72,73,76), these techniques currently are primarily limited to experimental imaging in small animals.

Near-infrared fluorescence imaging relies on light with a defined bandwidth as a source of photons that encounter a fluorescent molecule (optical contrast agent), which emits a signal with different spectral characteristics that can be resolved with an emission filter and captured with a high-sensitivity charge-coupled-device camera. We have recently developed autoquenched near-infrared fluorescent probes that become detectable after enzyme (eg, protease) activation, and we have shown that this quenching and de-quenching strategy represents an amplification strategy that increases target-to-background ratios over several hundred fold (54). Cell culture, *in vitro*, and *in vivo* study results have confirmed that the probes had very low fluorescence unless activated by proteases and were detectable in nanomolar amounts *in vivo* with no apparent toxicity at concentrations tested. Imaging of cathepsin B and cathepsin H protease activity allowed the detection of submillimeter-sized tumors (Fig 8) (12). More recently, using cathepsin D-negative and cathepsin D-positive mouse tumor models, we have also shown the power and specificity of this imaging approach for imaging certain transgenes in mice (79). Most important, however, the activatable near-infrared fluorescence approach holds promise for imaging of a number of endogenous proteases involved in cancer, infection, inflammation (including inflammatory arthritis), cardiovascular disease, and degenerative diseases in humans.

Another approach to imaging of endogenous gene products with optical techniques is by using fluorescent proteins. Specifically, imaging of green fluorescent protein can be performed in a manner analogous to that of near-infra-

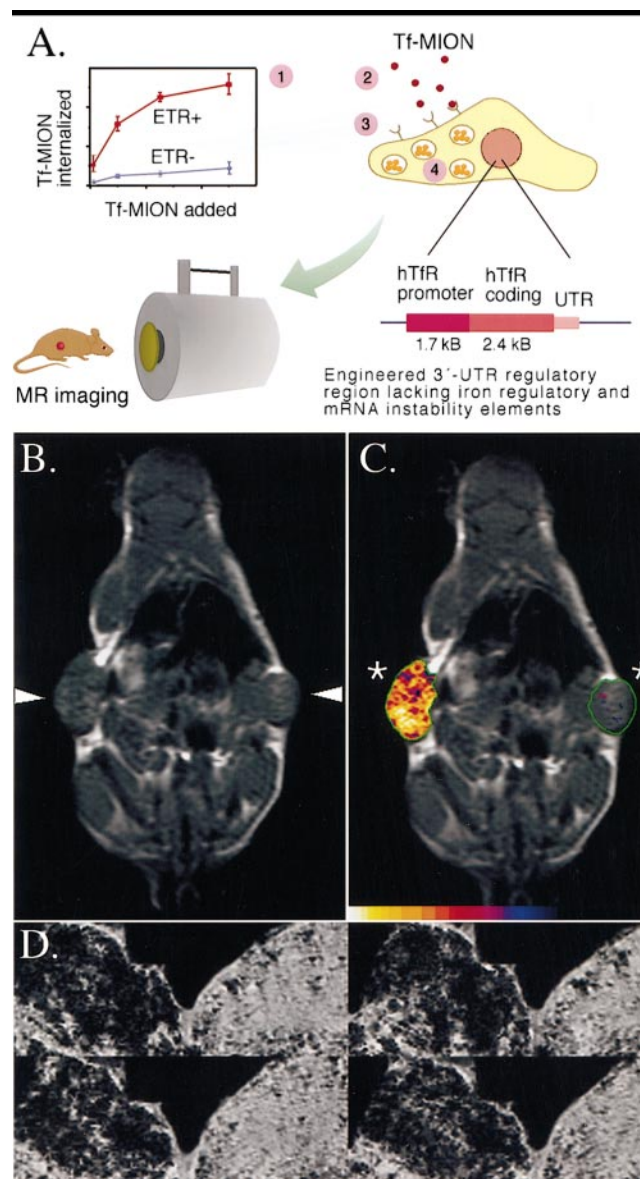


Figure 7. MR imaging of gene expression. *ETR* = engineered transferrin receptor, *hTfR* = human transferrin receptor, *mRNA* = messenger RNA, *Tf-MION* = transferrin-monocrystalline iron oxide nanoparticles, *UTR* = untranslated region. **A**, Schematic illustrates multiple amplification strategies to visualize gene expression with MR imaging: receptor overexpression, use of superparamagnetic substrates, receptor recycling, and increase in relaxivity by compartmentalization. **B**, Typical coronal spin-echo MR image (repetition time msec/echo time msec, 300/11; 1.5 T) of a mouse with engineered transferrin receptor-positive (left arrowhead) and engineered transferrin receptor-negative (right arrowhead) flank tumors. **C**, Composite of a coronal T1-weighted spin-echo MR image (image in **B** with color information superimposed) with superimposed R2 changes after transferrin-monocrystalline iron oxide nanoparticles administration, displayed as a color map. Note the difference in signal intensity between engineered transferrin receptor-negative (left *) and engineered transferrin receptor-positive (right *) tumors. **D**, Four contiguous 39- μ m-thick gradient-echo MR images (150/3.5, 35° flip angle, 7.1 T) of engineered transferrin receptor-positive tumor (left) and engineered transferrin receptor-negative tumor (right). Note the marked difference in tumor MR signal intensity in these otherwise matched 9L tumor pairs. (Base of each square is approximately 1 mm.) (Reprinted, with permission, from reference 13.)

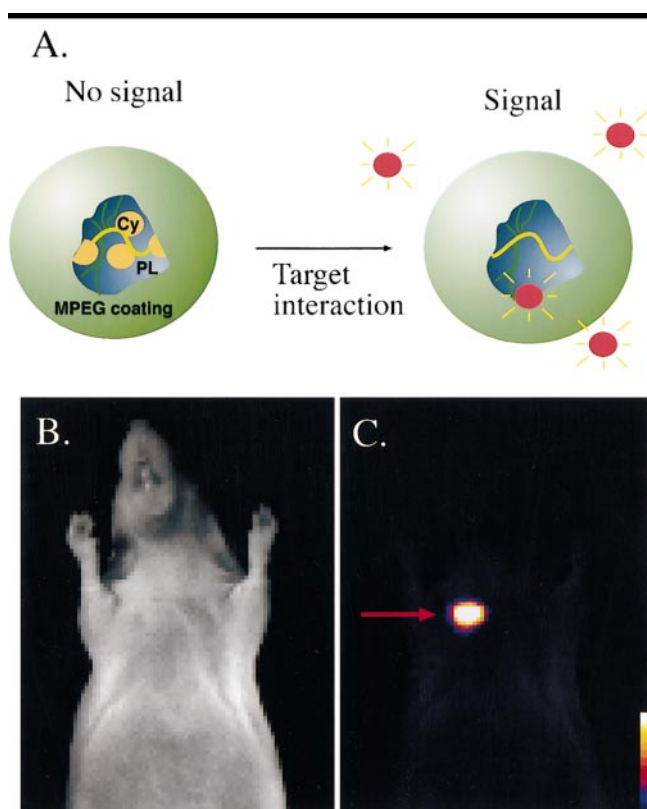


Figure 8. A, Schematic of near-infrared fluorescence probe activation. The initial proximity of fluorochrome molecules to each other results in signal quenching. After protease activation, fluorochromes become detectable (lightbulb effect). Cy = cyanine fluorochrome, MPEG = methoxy-polyethylene glycol, PL = poly-L-lysine. B, Light image of LX1 tumor implanted into the mammary fat pad of a nude mouse. The tumor is not detectable. C, False-colored near-infrared fluorescent image superimposed on the white light image shows cathepsin B/H enzyme activity, which allows the detection of this small tumor (arrow) in the mammary fat pad. (Reprinted, with permission, from reference 12.)

red fluorescence imaging with exception that (a) absorption and excitation (489 and 508 nm, respectively) are in the visible light range and (b) no exogenous fluorochromes need to be administered to enable visualization of green fluorescent protein expression. Given the wavelengths of excitation and emission, the technology is limited to surface structures (depth penetration, approximately 1–2 mm) in experimental animals. The authors of several elegant studies have used green fluorescent protein expression driven by promoters of interest to image protein expression in disease. For example, green fluorescent protein driven by the vascular endothelial growth factor promoter has revealed that a large amount of this factor in tumors is produced by surrounding stromal cells rather than by the tumor cells themselves (52).

Bioluminescence imaging exploits

the emission of visible photons at specific wavelengths based on energy-dependent reactions catalyzed by luciferases. Luciferase genes have been cloned from a large number of organisms, including bacteria, fireflies (*Photinus pyralis*), coral (*Renilla*), jellyfish (*Aequorea*), and dinoflagellates (*Gonyaulax*). In the firefly, luciferase utilizes energy from adenosine triphosphate to convert its substrate, luciferin, to oxyluciferin, with the emission of a detectable photon (53). Sensitive imaging systems have been built for the quantitative detection of small numbers of cells or organisms that express luciferase as a transgene (80). Expression of the bioluminescent reporter luciferase has specifically been used to image distribution and growth kinetics of transformed tumor cells (80) and bacterial organisms or the spatial distribution of gene expression products (53).

IMAGING OF OTHER MOLECULAR MARKERS AND PATHWAYS

Imaging of exogenous gene products after performance of gene therapy as described earlier is one of the many critical applications of molecular imaging. As the molecular mechanisms of many disease processes are dissected and better understood, additional imaging applications will emerge. In the following sections, we will discuss two specific areas of pathogenesis in which major advances in the understanding of molecular regulation have recently been made: angiogenesis and apoptosis. Through these specific examples, it will become apparent how molecular imaging techniques will dovetail with other developments. Reviews on other topics, including signal transduction (77), stem cell biology and cell-based therapies (82,83), antisense technology (84,85), cell cycle control (86,87), immunotherapy (31), and drug resistance (88), can be found elsewhere.

Angiogenesis

The term angiogenesis was first used to describe the growth of endothelial sprouts from preexisting postcapillary venules. More recently, the term has been used more broadly to encompass growth and remodeling of a primitive vascular network into a complex one. Angiogenesis occurs physiologically during embryonic development, the female reproductive cycle, wound healing, and hair growth. In adults (with the exception of the menstrual cycle), the vascular network is usually quiescent and regulated by tightly controlled angiogenic inducers and inhibitors. Under certain pathologic conditions, this balance is shifted, resulting in chaotic capillary growth. Up regulation and/or down regulation of the angiogenic process are central to approximately 30 diseases, including cancer, cardiovascular disease, immunologic disease, and diabetes. Understanding of the molecular basis of angiogenic control is slowly emerging (Fig 9), and excellent recent review articles exist on the topic (89–92). Key agonist stimulators of angiogenesis include vascular endothelial growth factor, basic fibroblast growth factor, angiopoietins, matrix metalloproteases, integrins, and cadherins, to name a few. Key endogenous inhibitors include thrombospondins, endostatin (20-kDa fragment of plasminogen), angiostatin (38-kDa fragment of plasminogen), troponin, metal-

lospodins, interleukins, and tissue inhibitors of matrix metalloproteinases, among others. While some of the latter biologic inhibitors are in early-phase clinical trials, there are also a number of small synthetic inhibitors in advanced clinical trials (for up-to-date information, see cancertrials.nci.nih.gov/news/angio/table.html). Overall, there are currently over 25 different antiangiogenic agents in phase 1 through phase 3 clinical trials.

Despite the continued development of additional angiogenesis inhibitors, a notable problem has been the need for surrogate markers to monitor drug effects. For example, chronic interferon administration for giant cell tumors of the mandible (93) or for life-threatening hemangiomas (94) only show treatment results after several months followed by a very gradual regression of these benign tumors. It has become clear that imaging of physiologic or molecular markers may be the most fruitful avenue yet to aid in the evaluation of response. A number of imaging techniques, in particular functional MR imaging and nuclear techniques that rely on first-pass or equilibrium contrast material enhancement, are available for studying microvascular circulation. For more in-depth information, a number of publications are available (95–99). Physiologic parameters that can easily be extracted include flow, perfusion, vascular volume fraction, permeability, and/or vascular structure.

A more specific approach to imaging angiogenesis is to target specific imaging probes to markers expressed on the altered neoendothelial surface of tumor vessels; that is, the same ones that represent targets for newly developed therapeutic drugs (Fig 9). The attractiveness of imaging and treating at the same molecular target is obvious and may allow the earliest target assessment possible. This is of particular importance, since phenotypic effects during therapy may not become apparent for weeks or months. Results from a number of pioneering imaging studies have targeted specific endothelial markers and attest to the feasibility of the approach. Specific examples include targeting of endothelial integrin $\alpha V\beta 3$ by using paramagnetic liposomes for MR imaging (100), targeting of the angiogenesis-associated fibronectin isoform by using optical probes (101); radiolabeled peptidomimetics of a urokinase plasminogen activator receptor antagonist (102), peptides adhering to the glycoprotein IIb/IIIa receptor on the surface of activated platelets, a major component of active thrombus formation (103);

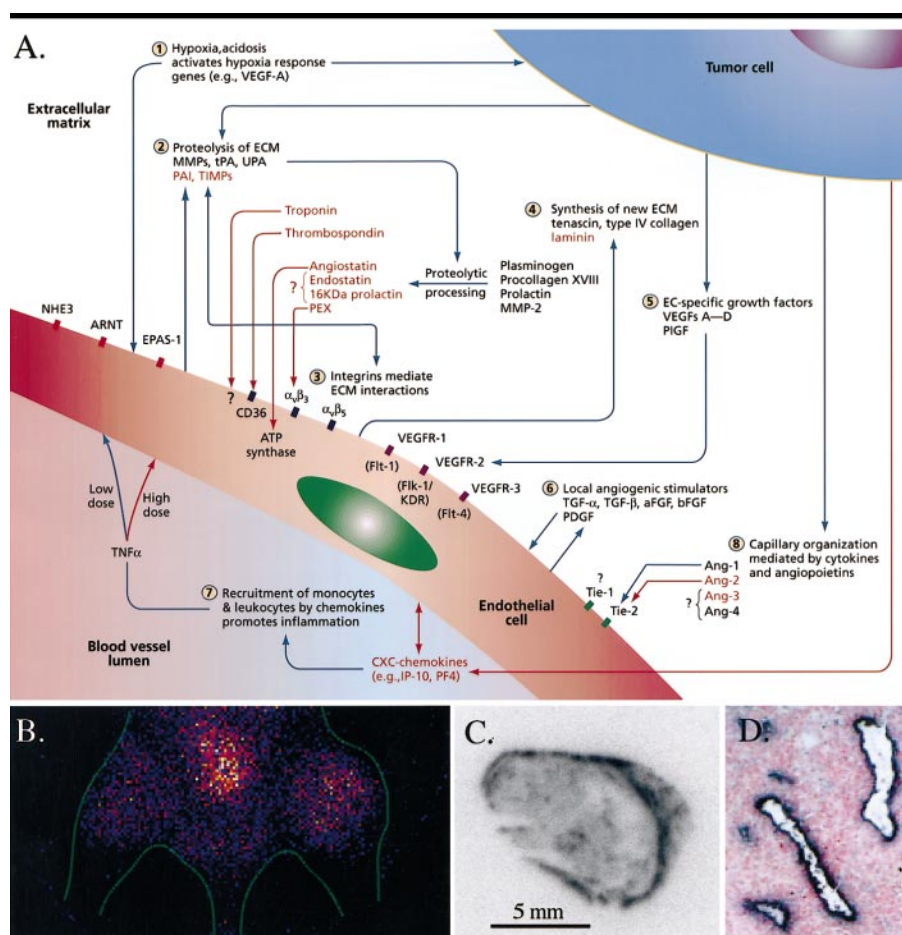


Figure 9. A, Schematic shows a simplified model of angiogenesis. Inhibitory molecules are shown in red. ATP = adenosine triphosphate, ECM = extracellular matrix, FGF = fibroblast growth factor, IP-10 = interferon-inducible protein 10, PAI = plasminogen activator inhibitor I, PDGF = platelet-derived endothelial cell growth factor, PEX = noncatalytic fragment of matrix metalloproteinase 2, PF4 = platelet factor 4, PlGF = placental growth factor, TGF = transforming growth factor, TIMP = tissue inhibitor of matrix metalloproteinases, $TNF\alpha$ = tumor necrosis factor α , tPA = tissue plasminogen activator, UPA = urokinase plasminogen activator, VEGF = vascular endothelial growth factor. (Reprinted, with permission, from reference 102.) B, Scintigram shows bilateral B16 tumors after intravenous injection of radiolabeled anti-transforming growth factor β antibodies. C, Autoradiograph shows an 8- μ m-thick frozen section of a B16 tumor obtained 45 minutes after intravenous injection of the antibody. D, Photomicrograph shows B16 tumor section after immunohistologically staining for the intravenously injected monoclonal antibody and revealed with a secondary antibody. (Original magnification, $\times 160$.) (Reprinted, with permission, from reference 104.)

and antibodies against the tumor growth factor β receptor (Fig 9) (104).

Although these examples are encouraging, more experimentation is needed to validate these new imaging strategies and translate them into the clinic. Once validated, these approaches can be used to answer questions such as the following: Is administration of a single antiangiogenic agent sufficient? What is the best administration strategy? Do we deliver sufficient inhibitors to their targets? Do the inhibitors truly affect their target in vivo? While these questions are highly relevant in tumor treatments, therapeutic angiogenesis using vascular endothe-

lial growth factor, fibroblast growth factor, or their genes for ischemic heart disease and peripheral vascular disease are other applications that will also ultimately benefit from this type of imaging research.

Apoptosis

Apoptosis is a physiologic form of programmed cell death critical for organ development, tissue homeostasis, and removal of defective cells in vivo without causing a concomitant inflammatory response. Apoptosis depends on the recognition of multiple extracellular and

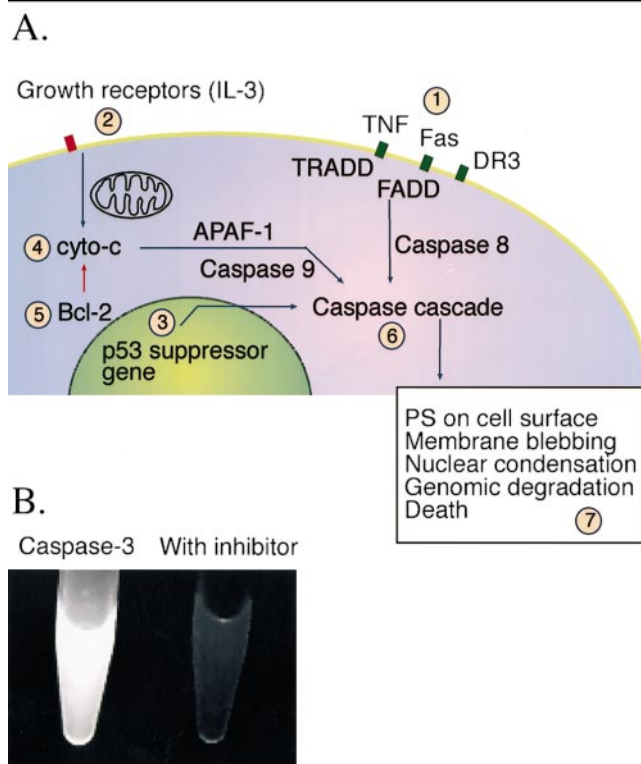


Figure 10. A, Schematic shows a simplified model of apoptosis. Three distinct pathways can result in apoptosis: activation of a death receptor (1), withdrawal of a growth factor (2), or DNA damage leading to a p53-induced apoptosis (3). The second pathway leads to release of cytochrome C from mitochondria (4) and is regulated by a family of Bcl-2 proteins (5). Once the caspase cascade involving the key caspase, caspase 3, is activated (6), cell death will automatically result (7). Caspase 3 represents an important imaging target (see text for details). During the terminal stages of apoptosis, intracellular phosphatidylserine (PS) is expressed on the cell surface and can be imaged with labeled annexin V (a high-affinity molecule for phosphatidylserine). B, Fluorescent probes for optical imaging that allow detection of caspase 3 activity (Tung CH, unpublished results, 2000). Note the markedly lower signal intensity when caspase 3 inhibitor is present.

intracellular signals, integration and amplification of signals, and activation of a family of effector proteases called caspases (the “henchmen”). Defects in control of apoptotic pathways may contribute to a variety of diseases. For example, apoptosis can be decreased (eg, in cancer, autoimmune disease, viral infections) or increased (eg, in AIDS, neurodegenerative disorders, ischemia, stroke, myelodysplastic syndromes).

There exist several excellent and detailed reviews (32,105–107) of the complex regulatory network and molecular mechanism that control apoptosis. Figure 10 summarizes the three classical pathways of apoptosis. The first pathway involves signaling of a death receptor (eg, Fas or tumor necrosis factor), which then activates caspases. A second classical pathway is initiated by the withdrawal of

growth factors (eg, interleukin-3), which leads to the release of cytochrome C from mitochondria followed by activation of caspases. This pathway is controlled by Bcl-2 family of proteins. The third pathway is induced by DNA damage (caused by anticancer drugs or radiation) that triggers p53 pathways of caspase activation. Once the apoptotic pathway is activated, caspase-mediated proteolysis is irreversible and ultimately leads to typical cellular changes, including cell shrinkage, membrane blebbing, chromatin condensation, and nuclear fragmentation.

The apoptosis pathways represent special opportunities for therapeutic intervention. Caspase inhibitors are attractive drugs for diseases in which apoptosis is increased, including ischemic reperfusion injury, neurodegenerative diseases, transplantation rejection, and autoim-

mune disorders. On the other hand, caspase activators, p53, or both may become useful strategies to induce more efficient cell killing in tumors. As these drugs are developed (32), it is critical to assess drug efficacy at the molecular level either by imaging caspase activity directly or by imaging other downstream effects such as annexin V binding to cells that abnormally express surface phosphatidylserine.

The authors of several excellent articles (108–110) have shown that radiolabeled annexin V can be used to image apoptosis in acute transplant rejection, in anti-Fas-induced fulminant hepatic apoptosis, and in cyclophosphamide treatment of murine lymphomas. In these studies, ^{99m}Tc -labeled annexin accumulated two- to sixfold higher in apoptotic tissues. While these results attest to the feasibility of imaging cells undergoing apoptosis (ie, disease detection), other imaging research is directed at measurement of intracellular caspase activity to potentially monitor caspase inhibitor treatments. Two approaches have been presented to date in which either caspase 3 peptide substrates (...DEVD... amino acid sequence, where D is aspartic acid, E is glutamic acid, and V is valine) that contain either nuclear labels (111) or near-infrared optical fluorochromes (Fig 10) were used. To internalize these molecular probes into cells, nuclear translocation signals such as Tat peptides (1) are generally used. Preliminary studies attest to the feasibility of measuring caspase 3 activity, but more extensive in vivo results are needed.

IMAGING OF GENOMICALLY MANIPULATED MICE

While the aforementioned examples highlight recent advances in molecular imaging, there has also been a recent interest in phenotypic imaging. Molecular geneticists looking for ways to model human diseases and companies testing new drugs are creating an unprecedented demand for transgenic and knockout mice (which have one or more of their 80,000 genes disabled). It is estimated that more than 25 million mice will be raised this year for experimental studies, accounting for over 90% of all mammals in research. This number is expected to increase by 10%–20% annually over the next decade. To what is this demand due? For one, it is created by the ability to fine tune genetic alterations of model mice. The low maintenance cost, fecundity, and genetic sim-

ilarity of mice to humans are other factors. Many human genes have a related mouse version, which makes it possible to gain insights into human disease. The number of transgenic or knockout mice already available is enormous, with a substantial increase expected in the future: In an effort to sequence the entire mouse genome, in analogy to the Human Genome Project, United States governmental funding agencies have awarded \$130 million through 2001 to begin sequencing this genome (for an update, see ray.nlm.nih.gov/genome/seq/MmHome.html). Similar efforts are underway to sequence the rat genome.

Rising mouse costs and the relative scarcity of some genetically engineered mice have been a major impetus for in vivo mouse imaging as an alternative to sacrifice and histologic processing. With each transgenic mouse costing routinely \$150 (and up to \$30,000 for certain breeder pairs), an entire mouse health care industry is forming. Major commercial breeders sold an estimated \$200 million worth of rodents in 1999; a greater demand will drive this number upward in the years to come. Last year, the National Cancer Institute awarded 19 groups at 30 institutions funding to create potentially thousands of new mutant and transgenic mice. In a similar effort, the National Cancer Institute has also recently funded several small-animal imaging resource programs across the country to speed up development of high-resolution imaging technology to harness this opportunity to study animal models of human disease.

Several excellent review articles (112–114) exist on genomic manipulation, including transgenic technology (direct injection of a gene of interest into a recently fertilized one-cell embryo) and knockout technology (injecting blastocysts with targeted embryonic stem cells). While the technology of producing these animals is beyond the scope of this review, more in-depth information and extensive lists and repertoires can be found on the Web (www.jax.org).

With regard to imaging of small animals, the size of the imaged object, the total volume that must be evaluated, the spatial resolution (voxel size) necessary for meaningful in vivo data, and the total time that may be dedicated to acquiring an image are quite different for a 20-g mouse than for a 70-kg person. Instrumentation requirements reflect these differences and can be exploited to maximize the information that can be obtained from small-animal models of

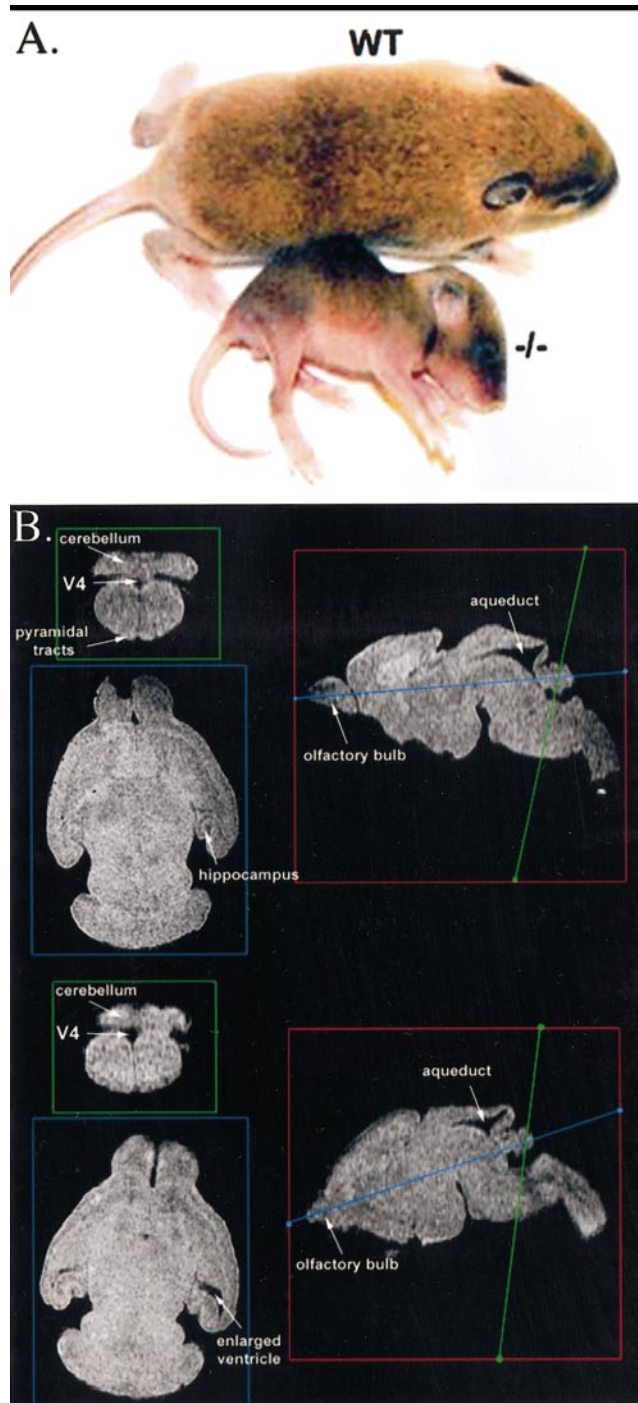


Figure 11. A, AnkyrinB (-/-) mouse and a normal littermate (WT). The normal littermate has two normal copies of the ankyrinB gene (ie, +/+). B, Micro-MR images of brains of normal littermate and ankyrinB mice. Parasagittal images of brains from a normal littermate (top right; WT in A) and an ankyrinB (bottom right; -/- in A) mouse. Left: MR sections defined by blue and green lines on images on the right are shown in their respective blue and green boxes. Although unmyelinated, the internal capsule and pyramidal tracts are clearly visible in the normal brain (top left) and are not evident in the mutant brain (bottom left). The internal capsule and corpus callosum are also visible on the parasagittal MR images of the normal brain (top right) and cannot be resolved on images of the mutant brain (bottom right). The lateral ventricles were abnormally enlarged in mutant brains (bottom left). V4 = fourth ventricle. (Reprinted, with permission, from reference 115.)

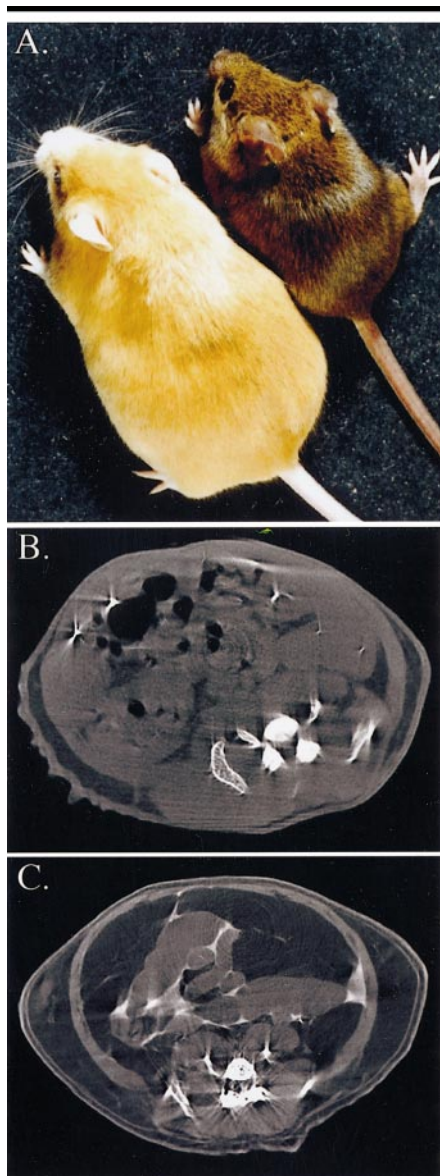


Figure 12. Micro-CT. Miniaturization of CT equipment allows imaging of transgenic and knockout mice at an in-plane resolution of 50 μm . *A*, In this example, a mutation of the agouti gene induces obesity in the mouse model, as manifested in increased body fat of one of the experimental animals (yellow coat). The other animal (brown coat) is a normal control mouse. *B*, Transverse CT image of a normal mouse. *C*, Transverse CT image of an obese mouse, as shown by amount of low-attenuation fat. (Courtesy of M. Paulus, MD, and E. Michaud, MD, Oak Ridge National Laboratories, Tenn.)

disease states. There has been a growing interest in dedicated rodent imaging systems in recent years, which the technological advances in the field demonstrate. The following is a brief summary of the state-of-the-art small-animal imaging systems and their primary applications. It

should be re-emphasized that phenotypic changes can occasionally be detected on the basis of physical parameters (eg, tumor development, physiologic abnormalities), while other abnormalities require molecular probes.

Micro-MR Imaging

MR imaging of rodents and analysis of specific rodent organs can be performed in narrow-bore (<20-cm-diameter) systems. The higher magnetic field and gradient strengths that can be achieved with these smaller bore sizes allow marked improvements in signal-to-noise ratio and spatial resolution. The micro-MR group at Duke University (Durham, NC) reports (115,116) routine acquisition of in vivo mouse brain images at resolutions of $50 \times 50 \times 500 \mu\text{m}$ in 2–3 hours (Fig 11). In vivo mouse cardiac imaging, with the associated problems of motion, has been reported at $120 \times 120 \times 1,000 \mu\text{m}$ resolution with a 7-T system (117,118), with a total imaging time in one case of 10–13 minutes (117). Ex vivo organ evaluation with MR imaging is somewhat easier because field and gradient strengths may be higher, dedicated hardware (radio-frequency coils) for the smaller samples results in a higher signal, physiologic motion (which can be a substantial problem with voxels on the order of tens of micrometers) is no longer an issue, and imaging times may be longer. As an example, $40 \times 40 \times 40\text{-}\mu\text{m}$ voxel images with a signal-to-noise ratio of about 30:1 were achieved in human brain hippocampal sections in about 11 hours (116). In vivo spatial resolution of $27 \times 16 \times 16 \mu\text{m}$ has been achieved in a 11.7-T magnet for imaging *Xenopus* embryos, with imaging times of approximately 4 hours (64). At higher field strengths (14.1 T), individual human progenitor cells have been imaged (1,119). To put this in perspective, typical clinical human brain MR voxels are $800 \times 800 \times 5,000 \mu\text{m}$.

Micro-CT

Although direct imaging of molecular events is somewhat limited by current computed tomographic (CT) techniques, micro-CT offers excellent spatial resolution of anatomy, which may be used to screen for phenotypes in transgenic mice and to evaluate the response to therapy. A resolution of 50 μm in the mouse has been achieved with an acquisition time of 15–20 minutes (120–122) (Fig 12). Ex vivo analysis of tissue specimens allows more favorable geometries and lack of physiologic motion. Resolution of $14 \times$

$14 \times 14\text{-}\mu\text{m}$ voxels (123) and smaller (124) have been reported for evaluation of trabecular bone samples.

Micro-PET and Micro-SPECT

Given the vast number of ligands and enzymatic precursors that can be radiolabeled, nuclear modalities are well suited for imaging of molecular events. The University of California–Los Angeles group has a developed a micro-PET system for rodents and small primates (125,126). The spatial resolution of the system is isotropic at 2 mm, equivalent to that reported by a number other PET device groups using different configurations and types of detectors. The Massachusetts General Hospital (Boston) PET group has constructed a single-ring PET imager with a spatial resolution of 1 mm (127, 128). Additional work on improved designs for small-animal PET devices, which take advantage of the geometry and size of mice, is being actively pursued. These improvements include substantially higher sensitivity, which some groups propose to achieve by surrounding the animal with detectors, and improved isotropic spatial resolution of less than 1 mm.

An alternative approach to PET that can be readily adapted to clinical systems is pinhole SPECT (129–131). While the sensitivity of SPECT systems is intrinsically about two orders of magnitude less than that of PET systems because of the need to collimate the input photons, the necessary radiopharmaceuticals and imaging systems are more readily available. In addition, higher doses per unit body weight can be given to rodents, as compared with the doses that can be administered in patients, which helps overcome some of the decrease in sensitivity. The spatial resolution of these systems has been reported to be as high as 1.7 mm. More recently, solid-state cadmium-zinc-telluride detectors scaled to the dimensions of mice have become available, allowing much higher sensitivity and, in combination with small-animal-specific grid design, similar or improved spatial resolution as compared with the pinhole designs. Ex vivo digital autoradiography, with routine spatial resolution as high as 25 μm , has also been used as an adjunct research tool to these in vivo nuclear technologies (Fig 6).

Micro-US

US imaging is a tradeoff between depth of penetration and spatial resolution, which are both related to the phonon

frequency used to evaluate tissue. As with other imaging modalities, the small size of a mouse can be used as an advantage in designing instrumentation. By using a transducer with a frequency in the 40–50-MHz range, imaging of mouse embryos in utero is possible at a 20–40- μ m transverse and 50–100- μ m lateral resolution (132). US, like CT, is primarily a tool for anatomic and physiologic imaging or for real-time intervention. Its role in molecular imaging includes evaluation of transgenic mice and image-guided transgene delivery, both of which may be performed in the early developmental or in utero phases (133). High-frequency Doppler US allows evaluation of the circulatory system. Blood flow in the heart and great vessels may be probed during mouse embryogenesis (134), and flow in vessels as small as 15–20 μ m in diameter may be sampled to evaluate microcirculation, for example, in angiogenesis (135,136).

Optical Imaging Techniques

There are a number of distinct approaches for probing tissue by using optical wavelengths. Phased-array (69,70) and diffuse optical tomographic (65,66,137) techniques have opened parts of the human body to imaging by means of near-infrared light by allowing imaging centimeters from the surface. At the microscopic level, 10- μ m resolution is achievable with optical coherence tomography (138,139), a technique analogous to US. Subsurface imaging of fluorescence (12,68) can be brought to the subcellular level in mouse models with intravital microscopy (52,77,78). Bioluminescence imaging (80,140,141) can be used to follow transgene expression in the whole animal, albeit at somewhat lower resolution. Overall, the size of the mouse is advantageous, especially in light of the mean scattering and attenuation lengths that are on the order of 1 mm and 1 cm, respectively, for near-infrared photons. The more invasive properties of optical coherence tomography and intravital microscopy, with their improved resolutions can also be exploited to help evaluate disease in mouse model systems.

IMPACT OF MOLECULAR IMAGING ON RADIOLOGY

To better understand the impact of the molecular revolution on medicine in general, it is helpful to look at the astonishing developments in other specialties. The most striking examples have been

the constant introduction of continually more specific drugs and inhibitors, the ability to genetically engineer and repair cells and tissues, and the commercialization of a multitude of diagnostic tools and kits (eg, DNA assays, protein assays). On the research side, an entire industry has developed that is aimed at speeding up molecular research and drug discovery. These tools have led, in part, to dramatic advances in the understanding of the molecular pathways of disease. The knowledge of new targets and pathways can now be efficiently translated into new drugs. We expect that these self-feeding development loops will ultimately allow the earlier detection of disease and more efficient treatments, resulting in improved patient care. Virtually all diseases will ultimately be affected by this molecular revolution. The question is not whether radiology should be involved, but how can we be certain that radiology is actively engaged in mainstream molecular medical research and in the translation of discoveries into clinically useful tools.

Radiology research in the past has been heavily focused on biomedical engineering and computing, and this engagement has served our specialty well. Today's imaging systems are markedly improved in terms of resolution and speed as compared with those of only a few years ago. Indeed, imaging technology is at a stage where imaging can occur at near-micrometer resolution, and concomitant acquisition of functional and physiologic information is possible. One of the key aspects of radiology's engagement in molecular imaging will be the recruitment of molecular biologists, cell biologists, and synthetic chemists into our specialty. These scientific specialties and the imaging specialties have long existed in separate worlds, which occasionally meet in uneasy partnerships. The bridge between these two worlds is difficult, but it is necessary to cross it to advance the field. In years to come, it will also be essential to attract and train the next cadre of imaging scientists to understand and comprehend mainstream molecular developments, actively engage in basic science research, and finally translate new discoveries into clinical applications.

Fortunately, the NIH leadership has clearly identified the aforementioned developments as opportunities for investment, and considerable funding is now available for molecular imaging research. This year alone, the National Cancer Institute has allocated an additional \$20 million to establish multiple centers for

molecular imaging and small-animal imaging resources across the country with the mission to explore novel in vivo imaging strategies (expected 5-year funding is more than \$70 million). This type of funding has occurred in addition to other available NIH and non-NIH federal funding sources. It appears that with these resources at hand, we have the opportunity to leap forward as a specialty. Ultimately, this research is expected to lead to vastly improved disease detection and molecular target assessment in vivo.

Despite the likelihood that today's practicing radiologist will regard the foregoing as too esoteric to influence his or her practice, we must all recognize that this is the future of the specialty. The search for a new breed of molecular scientist and support for research in this area is the responsibility of us all. Finally, dissemination of research results within the specialty and our departments will be critical to heighten the awareness of this emerging field.

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